

Nathan Holt

MATHEMATICIAN · SOFTWARE DEVELOPER · RESEARCHER · IMAGE SCIENCE ENGINEER

☎ (317) 412-3404 ✉ nathanwayneholt@gmail.com 🌐 nathanwayneholt.com

EDUCATION

Rochester Institute of Technology

Cumulative GPA: **3.94** / 4.00

M.S. IN APPLIED AND COMPUTATIONAL MATHEMATICS

Rochester Institute of Technology

Cumulative GPA: **3.32** / 4.00

B.S. IN COMPUTATIONAL MATHEMATICS

EXPERIENCE

L3Harris Technologies

Rochester, NY

TECHNICAL LEAD

May 2023 -

- Led a team of L3Harris engineers and sub-contractors from academia and industry to accomplish tasking and R&D goals on schedule and under budget.
- Briefed internal and external partners in both the Rochester, NY and Washington, DC areas.
- Led the creation of a software framework to model human decision-making and the methods of travel for large-scale synthetic GPS datasets used to train experimental neural network methods for detecting “anomalous” human behavior hidden amongst other normal mobility patterns.
- Supported the development and research of artificial intelligence methods to both detect deviations from normal human behavior in GPS datasets, and how best to “hide” anomalous human behavior in GPS datasets to avoid detection from other algorithms, in the style of a GAN.
- Supported and briefed IARPA and Intelligence Community members.
- Maintained an active TS/SCI clearance.
- Software development in Python, Rust, C++, Java.

L3Harris Technologies

Rochester, NY

IMAGE SCIENCE ENGINEER

Jan. 2019 - May 2023

- Became a lead software developer on several key software libraries that have the following general purposes:
 - big data analytics of human mobility datasets to provide actionable intelligence
 - software to generate artificial human behavioral patterns and simulate the movement of artificial humans over a given region of interest for exploitation purposes
 - automation software to generate synthetic imagery based on physics using ray-tracing
 - image processing and exploitation methods for remote sensing data
- Improved machine learning and artificial intelligence capabilities for object tracking and Markov prediction in imagery.
- Supported and briefed the NGA and other Intelligence Community members.
- Maintained an active TS/SCI clearance.
- Software development in Python, C++.

Harris Corporation

Rochester, NY

SOFTWARE ENGINEER INTERN

May 2018 - Aug. 2018

- Developed a software toolkit to analyze large-scale datasets of GPS data from object tracking algorithms.
- Applied knowledge of mathematics, data science, statistical physics, and spectral graph theory to develop experimental metrics that gave insight into customer-specified areas of interest.
- Created technical documentation and gave talks to non-technical colleagues to illustrate use of software and metrics and help reach a larger audience.
- Software development in Python.

SKILLS

Programming Languages Python, Rust, SQL, C++, R, MATLAB, HTML, CSS, C#, Java, PHP

Developer Tools Git, Atlassian, AWS, Docker, GitHub, GitLab

Key Interests Modeling, Simulation, Geospatial Science, Activity-Based Intelligence, Machine Learning

Mathematics Dynamical Systems, Differential Equations, Mathematical Modeling, Graph Theory